MEANING OF RESEARCH

Research in common parlance refers to a search for knowledge.

a scientific and systematic search for pertinent information on a specific topic.

a careful investigation or inquiry specially through search for new facts in any branch of knowledge

Research is an academic activity and as such the term should be used in a technical sense.

According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organising and evaluating data; making deductions and reaching conclusions; and at last carefully testing the conclusions to determine whether they fit the formulating hypothesis.

Research Process

1) Formulating the research problem;
2) Extensive literature survey;
3) Developing the hypothesis;
4) Preparing the research design;
5) Determining sample design;
6) Collecting the data;
7) Execution of the project;
8) Analysis of data;
9) Hypothesis testing;
10) Generalisations and Interpretation, and
11) Preparation of the report or presentation of the results, i.e., formal write-up of conclusions reached.

OBJECTIVES OF RESEARCH

1. **To gain familiarity with a phenomenon** or to achieve new insights into it (studies with this object in view are termed as exploratory or formulative research studies);

2. **To portray accurately the characteristics** of a particular individual, situation or a group (studies with this object in view are known as descriptive research studies);

3. **To determine the frequency with which something** occurs or with which it is associated with something else (studies with this object in view are known as diagnostic research studies);

4. **To test a hypothesis of a causal relationship between variables** (such studies are known as hypothesis-testing research studies).

TYPES OF RESEARCH

On the basis of Purpose

- **Descriptive vs. Analytical**: Descriptive research includes surveys and fact-finding enquiries of different kinds.
- The main characteristic of this method is that the researcher has no control over the variables; he can only report what has happened or what is happening. For example, frequency of shopping, preferences of people, or similar data.
- In **analytical** research, on the other hand, the researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material.

Exploratory Research

- **Purpose:** To explore a relatively unknown topic or problem — the "what could be."
  Goal: Gain insights, generate ideas, or identify patterns for further research.
  Often used when there is little existing literature.
- **Common methods:** Focus groups, interviews, open-ended surveys, pilot studies, literature reviews.

Examples:

- Interviewing teachers to explore **the challenges of using AI tools in classrooms**.
- Conducting focus groups to understand **why young adults are losing interest in traditional media**.
- A preliminary study exploring **potential health impacts of a new dietary supplement**.
- Reviewing literature to identify **emerging trends in renewable energy adoption**.

Descriptive Research

Purpose: To describe characteristics of a population or phenomenon — the "what is."
Goal: Provide an accurate picture of current conditions without investigating cause-and-effect.

- **Common methods:** Surveys, observations, case studies, document analysis.

Examples:

- A survey describing the **average age, income, and education level of smartphone users** in a city.
- A study documenting the **nutritional habits of high school students**.
- An analysis of **crime rates in different neighborhoods** over the past year.
- A report on **customer satisfaction levels** with a new public transportation system.

Analytical (or Explanatory) Research

- **Purpose:** To analyze relationships between variables or test hypotheses — the "why" and "how."
  Goal: Explain causes, correlations, or mechanisms behind observed phenomena.
- **Common methods:** Experiments, statistical analyses, regression, comparative studies.

Examples:

- Analyzing whether **students who use spaced repetition apps perform better on exams** than those who don't.
- A regression study examining **how income and education levels affect political participation**.
- Testing whether **a new teaching method improves math achievement** compared to traditional methods.
- Comparing **sales data before and after a marketing campaign** to determine its effectiveness.

Based on Application

Applied vs. Fundamental

- **Applied research(Action)** aims at finding a solution for an immediate problem facing a society or an industrial/business organisation, whereas **fundamental research(pure/basic)** is mainly concerned with generalisations and with the formulation of a theory.
- Basic research is directed towards finding information that has a broad base of applications and thus, adds to the already existing organized body of scientific knowledge.

Developing the Hypothesis

A hypothesis is a proposed statement that is testable and is given for something that happens or observed.

Ex: Studying more can help you do better on tests.

Null Hypothesis (H0)

There's no connection or difference between different things.

- *The average test scores of Group A and Group B are not much different.*
- *There is no connection between using a certain fertilizer and how much it helps crops grow.*

Alternative Hypothesis (H1 or Ha)

Alternative Hypothesis is different from the null hypothesis and shows that there's a big connection or gap between variables.

Scientists want to say no to the null hypothesis and choose the alternative one.

Example:

- *Patients on Diet A have much different cholesterol levels than those following Diet B.*

Type I and Type II Errors

Type I and Type II Errors are central for hypothesis testing.

False discovery refers to a **Type I error** where a true Null Hypothesis is incorrectly rejected.

On the other end of the spectrum, **Type II errors** occur when a false null hypothesis fails to get rejected.

Type I error, also known as a **false positive**, occurs in statistical hypothesis testing when a null hypothesis that is actually true is rejected.

Type II error, also known as a **false negative**, occurs in statistical hypothesis testing when a null hypothesis that is actually false is not rejected.

Minimizing Type I Error

- To reduce the probability of a Type I error (rejecting a true null hypothesis), one can choose a smaller level of significance (alpha) at the beginning of the study.

Minimizing Type II Error

- The probability of a Type II error (failing to reject a false null hypothesis) can be minimized by increasing the sample size

Significance Level

The **significance level**, usually written as **α (alpha)**, is a number used in **hypothesis testing** in statistics.

- Alpha is the **probability of making a Type I error**, which means:
- **Rejecting the null hypothesis when it is actually true**

Typical choices for α are:

- **0.05** → 5% risk (most common)
- **0.01** → 1% risk (more strict)
- **0.10** → 10% risk (less strict)

How it's used

- You **choose α before** doing the test.
- You calculate a **p-value** from your data. The *p* value, or probability value, tells you the statistical significance of a finding.
- Compare:
  - If **p-value ≤ α** → reject the null hypothesis, results are statistically significant.
  - If **p-value > α** → fail to reject the null hypothesis and **results are not statistically significant.**

Example

You design an experiment to test whether actively smiling can make people feel happier. To begin, you restate your predictions into a null and alternative hypothesis.

- H_0: There is no difference in happiness between actively smiling and not smiling.
- H_a: Actively smiling leads to more happiness than not smiling.

To test your hypothesis, you first collect data from two groups. The experimental group actively smiles, while the control group does not. Both groups record happiness ratings on a scale from 1–7.

- Next, you perform a *t* test to see whether actively smiling leads to more happiness. Using the difference in average happiness between the two groups, you calculate:
- a *t* value (the test statistic) that tells you how much the sample data differs from the null hypothesis,
- a *p* value showing the likelihood of finding this result if the null hypothesis is true.

- Through your hypothesis test, you obtain a *p* value of 0.0029. Since this *p* value is lower than your significance level of 0.05, you consider your results statistically significant and reject the null hypothesis.
- That means the difference in happiness levels of the different groups can be attributed to the experimental manipulation.

- **Power of a Test**

The **power** of a statistical test is the probability of **correctly rejecting the null hypothesis** when it is false.

Mathematically:

$$\text{Power} = 1 - \beta$$

where β is the probability of a **Type II error**.

Type II Error (β)

A Type II error occurs when we **fail to reject the null hypothesis** even though it is false.

Relationship:

$$\text{Power} = 1 - \beta \implies \beta = 1 - Power$$

As **power increases**, β must **decrease**, because the test is better at detecting a real effect.

Effect Size

While **statistical significance** shows that an effect exists in a study,

practical significance shows that the effect is large enough to be meaningful in the real world.

Statistical significance is denoted by ***p* values**, whereas **practical significance** is represented **by effect sizes.**

It's necessary to report effect sizes in research papers to indicate the practical significance of a finding.

The APA guidelines require reporting of effect sizes and confidence intervals wherever possible.

How do you calculate effect size?

The most common effect sizes are **Cohen's *d* and** **Pearson's *r***.

Cohen's *d* measures the size of the difference between two groups while

Pearson's *r* measures the strength of the relationship between two variables.

Preparation of Research Design

A **research design** is a strategy for answering your research question using empirical data. Creating a research design means making decisions about:

- Your overall research objectives and approach
- Whether you'll rely on primary research or secondary research
- Your sampling methods or criteria for selecting subjects
- Your data collection methods
- The procedures you'll follow to collect data
- Your data analysis methods

Descriptive Statistics

- Measures of Central Tendency (such as mean, median, mode)
- Measures of Dispersion (such as range, variance, standard deviation)
- Distribution shape (including skewness and kurtosis)

When the data consist of numerical scores that have been measured on an interval or ratio scale, construction of which frequency distribution graph would be suitable;

(a) Polygons

(b) Pie Chart

(c) Bar diagram

(d) All of above

1. When the data consist of **numerical scores measured on an interval or ratio scale**, we must choose a graph that is appropriate for **continuous numerical data**

(a) Polygons

Explanation:

Frequency Polygon is designed for **continuous or interval/ratio data**. It connects midpoints of class intervals with straight lines, which is appropriate because interval and ratio scales have equal units and a meaningful order.

Pie Chart (option b) is used for **categorical data**, not for continuous numerical scores. It shows proportions of categories, not frequencies of numerical intervals.

2. The purpose of z scores or standard scores is to identify and describe;

(a) Power of each score in a distribution

(b) Exact location of each score in a distribution

(c) Hypothetical value of each score

(d) All of above

3)The degrees of freedom determine the number of scores in the sample that are;

(a) Independent and free to vary

(b) Independent and located generally in

(c) Assume middle scores as zero

(d) Interdependent and poses restrictions

Degrees of Freedom (df): We use degrees of freedom in almost all inferential statistical tests.

- Degrees of freedom refer to the number of values in a calculation that are **free to vary** while still satisfying a given constraint (usually the sample mean).
- **Example:**
  Suppose you have a sample of 5 scores: 4, 6, 7, 8, and 5.
  The sample mean is 6.
  If you know **4 of the scores** and the mean, the **5th score is fixed** to maintain that mean.
  So, only **4 scores are "free to vary"**, which is why df = n – 1 = 5 – 1 = 4.

3. What would be the X-score for a population where z-score is = 2.00, $\mu = 50$ and $\sigma = 12$

- (a) 47
- (b) 74
- (c) 42
- (d) 24

- To find the **X-score** from a **z-score**, we use the formula:

$$X = \mu + (z \cdot \sigma)$$

Given:

$$z = 2.00$$
$$\mu = 50$$
$$\sigma = 12$$

4. The sample outcomes that are very unlikely to occur if the treatment has no effect is known as;

(a) The critical region

(b) Convincing evidence

(c) The alpha level

(d) The outlier scores

Critical Region (Rejection Region):

In hypothesis testing, the **critical region** refers to the **set of sample outcomes that are very unlikely to occur if the null hypothesis is true** (i.e., if the treatment has no effect).

If a test statistic falls in this region, we **reject the null hypothesis**.

Option b (Convincing evidence): Not precise. The term “convincing evidence” is informal; the technical term is **critical region**.

Option c (Alpha level): The **alpha level (α)** is the probability threshold we set (e.g., 0.05), not the outcomes themselves.

Option d (Outlier scores): Outliers are extreme data points, but not necessarily defined by hypothesis testing.

As the power of a test increases, the probability of a type-II error;

(a) Increases

(b) Unaffected

(c) Decreases

(d) None of the above

When a test is applied to two equivalent groups of subjects and the two sets of scores are matching, we get;

(a) cross-validation

(b) test-retest reliability

(c) equivalent form reliability

(d) rational equivalence reliability

Reliability and Validity of data

Reliability refers to how consistently a method measures something.

Validity refers to how accurately a method measures what it is intended to measure.

Types of Reliability

1. Equivalent Form Reliability (Alternate Form Reliability):

This type of reliability is used when **two equivalent forms of a test** are administered to **two equivalent groups** (or the same group at different times).

If the **scores match closely**, it indicates that the test has **high reliability**, meaning both forms measure the same construct consistently.

Interrater reliability	The consistency of a measure **across raters or observers**: do you get the same results when different people conduct the same measurement?
Internal Consistency	The consistency of **the measurement itself**: do you get the same results from different parts of a test that are designed to measure the same thing?

Types of Validity

Construct validity	The adherence of a measure to **existing theory and knowledge** of the concept being measured.
Content validity	The extent to which the measurement **covers all aspects** of the concept being measured.
Criterion validity	The extent to which the result of a measure corresponds to **other valid measures** of the same concept.

a) Cross-validation: Refers to testing the predictive accuracy of a test
on a **different sample**, not reliability per se.
(b) Test-retest reliability: Involves giving **the same test** to
the **same group at two different times**, then correlating scores.
(d) Rational equivalence reliability: This is not a standard term in psychometrics; likely a distractor.

In correlation research, the variables can be controlled by;

(a) Experimenter's manipulation and observation

(b) Control through observation

(c) Selection and statistical control

(d) Experimenter's manipulation and Statistical

Correlation Research and Control of Variables:

In **correlational research**, the researcher **does not manipulate variables**; instead, they **observe relationships as they naturally occur**.

Since variables aren't manipulated, control must be achieved using **other methods**:

Selection Control:

- Selecting participants or groups to **minimize extraneous variables**.
- Example: Matching participants on age or gender to reduce confounding effects.

Statistical Control:

- Using statistical techniques (e.g., partial correlation, regression) to **remove or account for the influence of extraneous variables**.

Why not the other options?

(a) Experimenter's manipulation and observation: Incorrect—correlational research **does not manipulate variables**.

(b) Control through observation: Observation alone **cannot control confounding variables**.

(d) Experimenter's manipulation and statistical: Manipulation is not part of correlational research.

2. **Z-scores (Standard Scores):**

- A **z-score** tells us **how many standard deviations a particular score is above or below the mean**. It **standardizes** scores from different distributions so they can be compared.
- **Purpose:** To **identify the exact location of a score within its distribution**.
  - A positive z-score means the score is above the mean.
  - A negative z-score means the score is below the mean.
  - A z-score of 0 means the score is exactly at the mean.

•**Option a (Power of each score):** Not correct.
Z-scores do not indicate "power" of a score; they indicate **relative position**.
•**Option c (Hypothetical value of each score):** Not correct.
Z-scores are **derived from actual observed values**, not hypothetical values.
•**Option d (All of above):** Not correct because only option **b** is accurate.

Research Approaches

Quantitative approach and the Qualitative approach

Generation of data in quantitative form which can be subjected to rigorous quantitative analysis in a formal and rigid fashion.

- Inferential
- Experimental
- Simulation

Which among the following cannot be obtained through ordinal data?

(a) Mean

(b) Median

(c) Mode

(d) Standard Error

Types of Scales

There are four different scales of measurement. The data can be defined as being one of the four scales. The four types of scales are:

- Nominal Scale
- Ordinal Scale
- Interval Scale
- Ratio Scale

Ordinal Data:

Ordinal data **represents categories with a meaningful order**, but the **intervals between ranks are not equal or known**.

Examples: Rankings (1st, 2nd, 3rd), Likert scales (agree, neutral, disagree).

Median: Can be used, because it only depends on the **order of data**.

Mode: Can be used, because it identifies the **most frequent category**.

Mean: Cannot be used meaningfully, because **calculating the average assumes equal intervals**, which ordinal data does not have.

Standard Error / Standard Deviation: Cannot be calculated properly for the same reason—they require **interval or ratio data**.

Key point: Standard Error **requires interval or ratio data**, because it relies on meaningful averages and deviations—something ordinal data **cannot provide**.

Nominal Scale

- A nominal scale is the 1st level of measurement scale in which the numbers serve as "tags" or "labels" to classify or identify the objects.
- Deals with the non-numeric variables or the numbers that do not have any value.
- There is no Order between the categories.

Example

What is your gender?

- M- Male
- F- Female
- Hair color: Black, Brown, Blonde
- Type of car: Toyota, Honda, Ford
- Ethnicity
- City of Birth

Ordinal Scale

- The ordinal scale is the 2nd level of measurement that reports the ordering and ranking of data without establishing the degree of variation between them.
- Ordinal represents the "order." Ordinal data is known as qualitative data or categorical data.
- It can be grouped, named and also ranked.

Example:

Education Level

- Categories: Primary < Secondary < Tertiary < Postgraduate
- You can **rank people by education**, but the "distance" between levels is not equal.

Socioeconomic Status

- Categories: Low, Middle, High

Customer Satisfaction

- Survey responses: Very Unsatisfied < Unsatisfied < Neutral < Satisfied < Very Satisfied
- **Likert Scales in Surveys**
- Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree

Interval Scale

The interval scale is the 3rd level of measurement scale.

- It is defined as a quantitative measurement scale in which the difference between the two variables is meaningful.
- In other words, the variables are measured in an exact manner, not as in a relative way in which the presence of zero is arbitrary.

Example:

- Temperature (Celsius or Fahrenheit)
- IQ (Intelligence Quotient) Scores
- Exam Test Scores

Ratio Scale

The ratio scale is the 4th level of measurement scale, which is quantitative. It is a type of variable measurement scale. **A true zero point**, which means "none" of the quantity exists.

One can categorize, rank and infer equal points between neighboring data points

Weight

- Example: 0 kg, 50 kg, 100 kg

Height

- Example: 0 cm, 150 cm, 180 cm

Income

- Example: $0, $50,000, $100,000

Age

- Example: 0 years, 20 years, 40 years

Distance

- Example: 0 km, 10 km, 20 km

Which method of sampling is generally preferred by qualitative researcher?

(a) Quota sampling

(b) Purposive sampling

(c) Snowball sampling

(d) Cluster sampling

Qualitative Research and Sampling:

In **qualitative research**, the focus is on **depth and richness of information**, not on generalizing to a population.

Researchers often **select participants intentionally** who are most likely to provide **insightful, relevant, or diverse perspectives** on the phenomenon being studied.

Purposive Sampling:

Also called **judgmental sampling**, it involves **selecting participants based on specific purpose or characteristics**.

Example: Interviewing only experienced teachers when studying innovative teaching methods.

Presenting someone else's ideas or words as one's own.

(a) Paraphrasing

(b) Plagiarism

(c) Pseudoscience

d) None of above

Plagiarism:

Definition: Using someone else's **ideas, words, or work** and presenting them as your own **without proper acknowledgment**.

It is considered **unethical and academically dishonest**.

Paraphrasing: Restating someone else's ideas in your **own words**, but you still need to **cite the source**. This is **not plagiarism if cited properly**.

Pseudoscience: Claims or practices that **appear scientific but lack scientific rigor**; unrelated to copying someone's work.

In a single subject research design, a series of observations of the same individual under the same conditions is called as;

(a) Treatment conditions

(b) Phase

(c) Interval

(d) All of above

Single-Subject Research Design:

In single-subject research, the focus is on **intensive study of one individual** over time.

Researchers often observe the participant under **different conditions**, usually including **baseline (no treatment)** and **treatment phases**.

Phase:

A **phase** is **a series of observations of the same individual under the same conditions**.

Example:

- **Baseline phase (A):** Observe behavior without intervention.
- **Treatment phase (B):** Observe behavior while applying the intervention.

By comparing data across phases, researchers can determine **the effect of the treatment**.

Follow-up hypothesis tests are done after an analysis of variance to determine exactly which mean differences are significant;

(a) Post hoc tests

(b) Fisher's ratio

(c) independent t-test

(d) None of above

Analysis of Variance (ANOVA) and Post Hoc Tests:

ANOVA tells us if there is a **significant difference among group means** overall.

However, it **does not indicate which specific groups differ** from each other.

To find out exactly **which pairs of means are significantly different**, we conduct **post hoc tests**.

Phi-coefficient is used when variables are;

(a) Dichotomous

(b) continuous

(c) Discrete

(d) All of above

Phi Coefficient (φ):

The **phi coefficient** is a measure of **association between two dichotomous (binary) variables**.

Dichotomous variables are those that take **only two possible values**, e.g., yes/no, male/female, pass/fail.

The phi coefficient is **essentially the Pearson correlation applied to 2 × 2 contingency tables**.

Research Methods versus Methodology

The scope of **research methodology** is wider than that of **research methods.**

Research methodology –Not only the **research methods** but also the logic behind the methods we use in the context of our research study and **why** we are using a particular method or technique and why we are not using others so that research results are capable of being evaluated either by the researcher himself or by others.

While **descriptive statistics** can only summarize a sample's characteristics, **inferential statistics** use your sample to make reasonable guesses about the larger population.

Since the size of a sample is always smaller than the size of the population, some of the population isn't captured by sample data. This creates **sampling error**, which is the difference between the true population values (called parameters) and the measured sample values (called statistics).

- A **statistic** is a measure that describes the sample (e.g., sample mean).
- A **parameter** is a measure that describes the whole population (e.g., population mean).
- Sampling error is the difference between a parameter and a corresponding statistic.

- There are two important types of estimates you can make about the population: point estimates and interval estimates.
- A **point estimate** is a single value estimate of a parameter. You survey 100 students and find their **average height = 165 cm**. It is more precise, less reliable.
- An **interval estimate** gives you a range of values where the parameter is expected to lie. A **confidence interval** is the most common type of interval estimate. It is less precise, more reliable.

Confidence Interval (CI)

We use a **confidence interval (CI)** whenever we want to **estimate a population parameter** (like a mean or proportion) based on a **sample**.

A **confidence interval (CI)** is a range of values, derived from sample data, that is likely to contain the true population parameter (like the mean or proportion) with a certain level of confidence.

Example:

- You survey 100 people and find that their **average height is 170 cm**, with a **95% confidence interval of 167–173 cm**.
- You are **95% confident** that the **true average height** of the entire population is **between 167 cm and 173 cm**.
- It **does not mean** there is a 95% chance that the true mean is in that range. Instead, **if you repeated this study 100 times**, about **95 of those intervals would contain the true mean**.

Formula (for a mean)

- $$\text{Confidence } Interval = \bar{x} \pm z\left(\frac{s}{\sqrt{n}}\right)$$

Where:

- $\bar{x}$ =sample mean
- z= z-score (from standard normal distribution; e.g., 1.96 for 95%)
- s= sample standard deviation
- n= sample size

- A **z-score** (also called a **standard score**) is a way of describing a score in terms of how far it is from the **mean** of a distribution, measured in **standard deviations**.
- A **positive z-score** means the score is **above the mean**, and a **negative z-score** means it is **below the mean**.

A z-score of **0** means the score is **exactly at the mean**.

z=X−μ/σ

X = raw score

μ= mean of the population

σ = standard deviation of the population

Reliability

Reliability of data refers to the **consistency or stability** of measurements.

Example: If a scale shows your weight as 70 kg every time you step on it, it is **reliable**. But if your actual weight is 68 kg, it is precise but **not accurate**.

Types of reliability

- **Test-retest reliability:** Same test given to the same group at two different times.
- **Equivalent form reliability:** Two equivalent forms of a test produce similar results.
- **Internal consistency:** Different items within a test measure the same construct consistently.
- **Inter-rater reliability:** Different observers or raters give similar scores.

Sampling

Sampling is a fundamental concept in statistics and research where we select a subset of individuals or items from a larger population to make inferences about the whole population.

Probability Sampling-Every member of the population has a known, non-zero chance of being selected

Non probability sampling**-not every member has a known chance of being selected**

Simple Random Sampling

- Each member of the population has an **equal chance** of being selected.

Student Survey

- **Population:** 500 students in a university.
- **Sampling:** Use a random number generator to select 50 students for a survey about campus facilities.

Employee Satisfaction

- **Population:** 200 employees in a company.
- **Sampling:** Draw 30 employee names randomly from the employee database to complete a satisfaction questionnaire.

Systematic Sampling

- Every *kth* member of the population is selected after a random starting point.
- Example: Selecting every 10th person on a list of 1,000 employees

Stratified Sampling

- Population is divided into **strata** (groups) based on a characteristic (e.g., age, gender), and a random sample is taken from each stratum.

Education Level Survey

- **Population:** All employees in a company.
- **Strata:** Education levels (High school, Bachelor’s, Master’s, PhD).
- **Sampling:** Randomly select a proportionate number of employees from each education level to survey their job satisfaction.

Healthcare Study

- **Population:** Patients in a hospital.
- **Strata:** Age groups (0–18, 19–35, 36–60, 61+).
- **Sampling:** Take a random sample from each age group to study the prevalence of a disease.

Market Research

- **Population:** Smartphone users in a country.
- **Strata:** Gender (Male, Female, Non-binary).
- **Sampling:** Randomly select a sample from each gender group to understand consumer preferences.

Cluster Sampling

Instead of dividing the population into strata based on a characteristic, the population is divided into **clusters (usually naturally occurring groups like schools, neighborhoods, or companies)**, and then a **random selection of entire clusters** is studied.

Health Survey

- **Population:** Residents of a state.
- **Clusters:** Hospitals or clinics.
- **Sampling:** Randomly select 20 hospitals and survey all patients visiting those hospitals.

Example

- Suppose you want to study the average math score of 5th graders in a large state.
- The state has **1,000 elementary schools** (clusters).
- Listing all 5th graders individually is impossible.

You could:

- Randomly select **50 schools (clusters)**.
- Survey **all 5th graders** (or a random sample of them) within those selected schools.

Non-Probability Sampling Methods

Convenience Sampling

- Selecting individuals who are easiest to reach.
- Example: Surveying people at a mall.

College Survey

- **Population:** All students at a university.
- **Sampling:** Survey students who happen to be in the library at the time.

Sample Size

- The "right" sample size depends on several factors: your **population size**, **research purpose**, **desired accuracy**, and **available resources**.
- Start by identifying your population.
- Let's say you're studying **higher education institutions in North India**.
- Population size (N): 293 universities (or 1,892 colleges)
- Decide on your confidence level and margin of error
- **Confidence level:** 95%

- The higher your confidence and the smaller your error, the **larger the sample** you'll need.
- A common formula for a finite population is:

n=N×Z2×p(1−p)/ e2(N−1)+Z2×p(1−p)

Where:

- n= sample size
- N= population size
- Z= z-score for confidence level (1.96 for 95%)
- p= estimated proportion of population attribute (use 0.5 if unknown)
- e= margin of error (e.g., 0.05)

Judgmental (or Purposive) Sampling

- Researchers select sample members **based on their judgment** of who will provide the best information.

Example: Interviewing experts in a field.

Healthcare Study

- **Population:** Patients with a rare disease.
- **Sampling:** Select patients who have detailed medical records and have been treated for the disease.

Market Research

- **Population:** Coffee drinkers in a city.
- **Sampling:** Choose individuals who regularly buy premium coffee for a taste-test survey.

Quota Sampling

Quota sampling is a **non-probability sampling method** where researchers select participants to ensure **specific groups (quotas) are represented**, but selection within each group is **non-random**.

Political Opinion Poll

- **Population:** Registered voters in a city.
- **Quota:** 40% Party A supporters, 35% Party B, 25% Party C.

Restaurant Customer Survey

- **Population:** Customers over a week.
- **Quota:** 50% dine-in, 50% takeout; 50% lunch, 50% dinner.

Snowball Sampling

- **Snowball sampling** is a **non-probability sampling technique** often used when the population is **hard to reach or hidden**.
- In this method, initial participants refer others, and the sample “snowballs” as more people are recruited through referrals.

Study on Entrepreneurs in Emerging Industries

- **Population:** Startup founders in a new technology sector.
- **Sampling:** Begin with a few known founders, then expand the sample through their professional networks.

1. Formulating the research problem

- There are two types of research problems, viz., those which relate to states of nature and those which relate to relationships between variables.
- The researcher must at the same time examine all available literature to get himself acquainted with the selected problem. He may review two types of literature—the conceptual literature concerning the concepts and theories, and the empirical literature consisting of studies made earlier which are similar to the one proposed.

Temperature (°C or °F)

- The difference between 20°C and 30°C is the same as between 30°C and 40°C (equal intervals).
- But 0°C ≠ “no temperature” — it’s just another point on the scale.
- You can’t say 40°C is “twice as hot” as 20°C.
- **No true zero point** — zero does **not** mean “none” of the quantity.

Frequency Distribution

A **frequency distribution** describes the number of observations for each possible value of a variable.

Frequency distributions are depicted using graphs and frequency tables.

Types of Variables in Research & Statistics | Examples

- Data is a specific measurement of a variable – it is the value you record in your data sheet. Data is generally divided into two categories:
- **Quantitative data** represents amounts
- **Categorical data** represents groupings

Quantitative variables are two kinds:

Discrete

Continuous

Discrete Variable

Variables that can take **specific, separate values** — usually **countable.**

- **Values**: Whole numbers (integers), no in-betweens.

Examples:

- Number of students in a class (you can have 25, not 25.5)
- Number of cars in a parking lot
- Number of goals scored in a match

Continuous Variables

Variables that can take **any value within a range** — often **measurable**.

• **Values**: Includes decimals/fractions.

Examples:

• Height (e.g., 170.2 cm)

• Weight (e.g., 62.8 kg)

• Time (e.g., 3.45 seconds)

• Temperature

The **independent variable** (the one you think might be the **cause**) and then measure the **dependent variable** (the one you think might be the **effect**) to find out what this effect might be.

Control variables: Variables that are held constant throughout the experiment.

Example: Drug Effect on Blood Pressure

• **Independent variable:** Type of drug

• **Dependent variable:** Blood pressure reduction

Control variables:

- Dosage amount
- Age and gender of participants
- Diet and exercise routine
- Time of day medicine is taken

Correlational research

- When you do correlational research, the terms “dependent” and “independent” don’t apply, because you are not trying to establish a cause and effect relationship (causation).
- There might be cases where one variable clearly precedes the other (for example, Study Hours and Exam Scores). In these cases you may call the preceding variable (i.e., the rainfall) the **predictor variable** and the following variable (i.e. the mud) the **outcome variable**

Confounding variables

A **confounding variable** is an **outside factor** that can:

Influence both the **independent variable** (what you manipulate)

- And the **dependent variable** (what you measure)
- …which can **confuse or distort** the results of a study.

- Confounding variables **make it hard to know** if the changes in your dependent variable were really caused by the independent variable — or by the confounder.
- A **confounding variable** is a type of **extraneous variable** that **varies systematically with the independent variable** and **affects the dependent variable**, making it difficult to determine the true relationship between the independent and dependent variables.

Example

-Does drinking coffee improve exam performance?

- **Independent variable (IV)**: Coffee consumption
- **Dependent variable (DV)**: Exam performance
- **Confounding variable**: Amount of sleep

If students who drank coffee also **slept better**, then better exam scores might be due to **better sleep**, not the coffee.
So sleep is a **confounding variable**.

Simple Definition:

- A **confounding variable** is a **hidden influencer** that affects your results and makes it hard to tell what's really causing what.

Latent variables

- **Latent variables** are variables that are **not directly observed** but are **inferred** from other variables that are observed (measured). They represent underlying, hidden constructs that influence the observed data.

Latent variables are like **hidden causes or traits** that we suspect are there because they affect what we can see.

For example:

- You can't directly **see** someone's "intelligence."
- But you can measure their test scores or reaction times.
- From those, you **infer** how intelligent they might be.

Composite variables

- A variable that is made by combining multiple variables in an experiment. These variables are created when you analyze data, not when you measure it.

Example: Socioeconomic Status (SES)

Components: Income, education level, occupational status

Composite Variable: SES = combined score of the three components.

Use: To measure a person's overall social and economic position.

Composite Variable Example

- Suppose you measure three things about students:
  - Attendance
  - Homework completion
  - Class participation
- During analysis, you **combine these three scores** into a single **"Engagement Score"**.
- This Engagement Score is **not measured directly** — it's **created from other variables**.

Calculate the $\sum X2$ value of the given scores: 6, 2, 4, 2.

$\sum X^2$ (the sum of the squares of the given scores).

Step 1: Square each score

- $6^2 = 36$
  $2^2 = 4$
  $4^2 = 16$
  $2^2 = 4$

Step 2: Add the squared values

- $\sum X^2 = 36 + 4 + 16 + 4$
  $\sum X^2 =$ **60**

A population has mean of μ=70 and a standard deviation of σ=5. If 10 point is added to every score in the population, what would be the new values for the population mean and standard deviation?

- When you add a constant value to every score in a population, the mean and standard deviation change in specific ways:
- **Mean (μ):** Adding a constant shifts the mean by that same amount.
- **Standard deviation (σ):** Adding a constant does **not** affect the spread, so the standard deviation stays the same.

Decision errors

Type I and Type II errors are mistakes made in research conclusions.

- A Type I error means rejecting the null hypothesis when it's actually true,

- Type II error means failing to reject the null hypothesis when it's false.

Degrees of Freedom

Degrees of freedom of the data set (total number of observations minus 1).

Simple Example

You have 3 test scores and know their **average** is 80:

Scores: ?, ?, ?

Average formula: (score1 + score2 + score3) ÷ 3 = 80 → total = 240

Now, if you know **the first two scores** are 70 and 90:

Third score = 240 – (70 + 90) = 80

Observation: Only **2 scores were free to vary**; the last one was **fixed by the average**.

- So, **degrees of freedom = 3 – 1 = 2**

Variance

- The **variance** is a measure of variability. It is calculated by taking the average of squared deviations from the mean.
- Variance tells you the degree of spread in your data set. The more spread the data, the larger the variance is in relation to the mean.
- The standard deviation is derived from variance and tells you, on average, how far each value lies from the mean. It's the square root of variance.

T-Distribution | What It Is and How To Use It (With Examples)

- The *t*-distribution, also known as Student's *t*-distribution, is a way of describing data that follow a bell curve when plotted on a graph, with the greatest number of observations close to the mean and fewer observations in the tails.
- It is a type of normal distribution used for smaller sample sizes, where the variance in the data is unknown.

QUESTION

After the recent heavy catastrophic floods in a given state, a psychologist interviewed 120 participants to understand the psychological consequences of the event. This is an example of

Ex-post facto field research-causal-comparative research

- **Ex-post facto research** means "after the fact." It studies events or phenomena that have already occurred, and the researcher cannot manipulate the independent variable.
- In this case, the **floods have already happened**, and the psychologist is studying their psychological impact **after the event** — no manipulation or control is possible.
- Because the interviews occur in a **real-world setting** (the field), it is also considered **field research**.

Field experiment: involves manipulating variables in a natural setting

- A psychologist wants to see whether **social support reduces stress** in disaster survivors.
- They select two similar groups of survivors.
- One group receives **a structured social-support program** (experimental group).

The other group does not (control group).

- After a few weeks, they measure both groups' stress levels.
- Here, the **independent variable (social support)** is **manipulated** in a **real-world setting**, making it a **field experiment**.

Controlled experiment: happens under laboratory conditions with manipulated variables

- A psychologist wants to test whether **sleep deprivation affects memory**.
- They recruit participants and randomly assign them to two groups:
  - Group A: stays awake for 24 hours.
  - Group B: gets a full night's sleep.
- Both groups then take the same memory test in a **lab setting**. Here, the **independent variable (sleep deprivation)** is **manipulated**, and the environment is **controlled** to eliminate other influences like noise, lighting, or distractions.

Action research: aims at solving an immediate problem or improving practices

A school counselor notices that students have high test anxiety.

- They **plan** and **implement** a relaxation-training program,
- **Observe** how it affects anxiety levels,
- **Reflect** on the outcomes, and
- **Modify** the program to improve results.
- Here, the research's main goal is **improving student well-being**, not just collecting data for theory — that's **action research**.

Power of a statistical test

- The **power of a statistical test** refers to the probability that the test will **correctly reject a false null hypothesis (H_0)**.

Mathematically,

- Power $= 1 - \beta$

where **β (beta)** is the probability of making a **Type II error** (failing to reject a false null hypothesis).

As power of test increases, probability of Type II error decreases.

Event sampling

- **Event sampling** is a method used mainly in **observational research**, where the researcher records specific events or behaviors **each time they occur** during a set observation period.
- This technique focuses on capturing **particular instances** or occurrences rather than continuous monitoring.

Ways of knowing or acquiring knowledge

- **The method of authority** refers to accepting information as true because it comes from a respected or authoritative figure, without questioning or doubting it.
- **The method of faith:** Relates more to belief without evidence, often tied to religious or spiritual beliefs, not necessarily authority-based.
- **The method of intuition:** Relies on personal feelings or instincts rather than external authority.

The Scientific Method: Systematic, evidence-based way of acquiring knowledge through observation, experimentation, and logical reasoning.

Research Methods

- **Randomization** is the process of using a random method (like a random number generator) to prevent any systematic relationship or bias between variables. It helps ensure that confounding variables are evenly distributed across groups.
- **Random assignment** is a specific type of randomization where participants are assigned to different groups or conditions randomly.
- **Sampling error** is the error that occurs when a sample does not perfectly represent the population.

- The **Phi-coefficient** is a measure of association used specifically when **both variables are dichotomous** (i.e., they each have two categories, like Yes/No, Male/Female).
- It is essentially the **Pearson correlation coefficient** applied to two binary variables.
- **Continuous:** For continuous variables, other correlation measures like Pearson's r are used.
- **Discrete:** Discrete variables with more than two categories generally require other measures like Cramér's V or contingency coefficients.

- **Post hoc tests** are conducted **after an ANOVA** when you find a significant overall effect, to pinpoint **exactly which group means differ** from each other.
- **Fisher's ratio:** Another name for the F-ratio used in ANOVA itself, not for follow-up tests.
- **Independent t-test:** Used to compare means between two groups, but not suitable for multiple comparisons after ANOVA without adjustments.

- In **single-subject research designs**, a **phase** refers to a period during which the same individual is observed under consistent conditions (e.g., baseline phase, treatment phase).
- **Treatment conditions:** Refers to the specific intervention applied, not the entire series of observations.
- **Interval:** Usually refers to time periods between observations, not the series of observations under the same condition.

- **Law of Comparative Judgment** was formulated by **Louis Leon Thurstone**.
- **Likert:** Known for the Likert scale (rating scales).
- **Stevens:** Known for Stevens' power law and levels of measurement.

Relaibility

- The **reliability of data** refers to how **consistent, stable, and dependable** the data is when measured or collected under similar conditions.
- Example: A bathroom scale that shows 70 kg today and 70.1 kg tomorrow is reliable. If it shows 70 kg today and 65 kg tomorrow, it’s not reliable.

- **Equivalent form reliability** (also called parallel-form reliability) refers to the consistency of test scores between two equivalent versions of the test administered to **two equivalent groups**.
- **Cross-validation:** Refers to testing a model on a new dataset to assess generalizability, not reliability between test forms.
- **Test-retest reliability:** Measures consistency of scores when the **same test** is administered to the **same group** at two different times.

- In **correlational research**, the researcher **does not manipulate variables** but observes relationships as they naturally occur.
- Variables are controlled by **selecting participants carefully** (selection control) and using **statistical methods** (like partial correlation, regression) to control for potential confounding variables.
- This differs from experimental research where the experimenter manipulates variables directly.

- **APA Style** (from the *American Psychological Association*) is a **complete writing and formatting system** used mainly in the social sciences (like psychology, education, sociology, business, etc.).
- **APA (American Psychological Association) guidelines**, the **abstract** in an APA-style paper should be:

Between 150 and 250 words.

- According to the **APA (American Psychological Association) guidelines**, a good research report should clearly include:
- **What was done** – A clear description of the research problem and objectives.
- **What procedure was followed** – Details about the methods, participants, materials, and procedure so the study can be replicated.
- **What was found** – Presentation of results, including statistical analyses and findings.
- **How the research is related to other knowledge** – Discussion of how the results fit into the broader context, implications, and limitations.

According to **APA guidelines**, a good research proposal should include:

- **What will be done** – A clear description of the research objectives and planned methods.
- **What may be found** – Expected outcomes or hypotheses based on theory or prior research.
- **What is the need** – The rationale for the study, explaining why it is important or necessary.
- **How your planned study is related to other research** – Discussion of the study's context within existing literature.

Research Paradigm

A research paradigm is a worldview or philosophical framework, including ideas, beliefs, and biases, that guides the research process.

3 pillars of Research Paradigm are

---**Ontology**- Study of nature of reality.

---**Epistemology**-study of knowledge and how we can know reality.

---**Methodology** is the study of how one investigates the environment and validates the knowledge gained.

Types of research paradigms

- Positivism: Knowledge is objective, observable, and measurable
- Interpretevism: Reality is socially constructed, and understanding human behavior requires interpreting the meanings and perspectives of participants.
- Humanistic: Focuses on the whole person, emphasizing personal growth, self-actualization, and human potential.
- Constructivism: Knowledge is constructed by individuals based on their experiences and interactions with the world.